

Project 3 — BJT Emitter Follower

1 Given Information and Design Goal

For the design:

- Source resistance: $R_S = 100\ \Omega$
- Load resistance: $R_L = 100\ \Omega$
- Input amplitude: $0.5\ \text{V}$
- Thermal voltage: $V_T \approx 25\ \text{mV}$

The two main requirements are:

1. Attenuation requirement: overall gain should satisfy

$$G \geq 0.9$$

2. Signal swing requirement: small-signal variation in v_{be} should remain small

The chosen topology is a common-collector (emitter follower) BJT amplifier.

2 Small-Signal Model

The small-signal emitter resistance is

$$r_e = \frac{1}{g_m}$$

Using

$$g_m = \frac{I_C}{V_T}$$

gives

$$r_e = \frac{V_T}{I_C}$$

Since for a BJT in active region,

$$I_E \approx I_C$$

we use

$$r_e \approx \frac{V_T}{I_E} \approx \frac{25\ \text{mV}}{I_E}$$

Also, the input resistance is approximated by

$$R_{in} = R_1 \parallel R_2 \parallel (\beta + 1)(r_e + R_L)$$

3 Attenuation Requirement

The overall gain is approximated by

$$G = \frac{R_{in}}{R_{in} + R_S} \cdot \frac{R_L}{r_e + R_L}$$

The design requirement is

$$G \geq 0.9$$

Substituting the known values $R_S = 100\ \Omega$ and $R_L = 100\ \Omega$:

$$\frac{R_{in}}{R_{in} + 100} \cdot \frac{100}{r_e + 100} \geq 0.9$$

To satisfy this expression, we want:

$$R_{in} \gg 100 \quad \text{and} \quad r_e \ll 100$$

As a first design target, choose

$$R_{in} \approx 10\ \text{k}\Omega \quad \text{and} \quad r_e \approx 1\ \Omega$$

Then

$$\begin{aligned} G &\approx \frac{10000}{10100} \cdot \frac{100}{101} \\ G &\approx 0.9901 \cdot 0.9901 \\ G &\approx 0.98 \end{aligned}$$

Therefore,

$$G \approx 0.98 > 0.9$$

so the attenuation requirement is satisfied.

4 Signal Swing Requirement

The signal swing requirement was written as

$$0.5 \cdot \frac{R_{in}}{R_{in} + R_S} \cdot \frac{r_e}{r_e + R_L} \leq 0.2x$$

where

$$x = 25\ \text{mV}$$

So the right-hand side becomes

$$0.2x = 0.2(25\ \text{mV}) = 5\ \text{mV}$$

Thus the condition is

$$0.5 \cdot \frac{R_{in}}{R_{in} + R_S} \cdot \frac{r_e}{r_e + R_L} \leq 5\ \text{mV}$$

Substituting $R_S = 100\ \Omega$ and $R_L = 100\ \Omega$ gives

$$0.5 \cdot \frac{R_{in}}{R_{in} + 100} \cdot \frac{r_e}{r_e + 100} \leq 5\ \text{mV}$$

Using the same design target as before,

$$R_{in} \approx 10 \text{ k}\Omega \quad \text{and} \quad r_e \approx 1 \Omega$$

we get

$$\begin{aligned} & 0.5 \cdot \frac{10000}{10100} \cdot \frac{1}{101} \\ &= 0.5 \cdot 0.9901 \cdot 0.00990 \\ &\approx 0.0049 \text{ V} \\ &\approx 4.9 \text{ mV} \end{aligned}$$

Hence,

$$4.9 \text{ mV} < 5 \text{ mV}$$

so the signal swing requirement is also satisfied.

This confirms that choosing

$$R_{in} \approx 10 \text{ k}\Omega \quad \text{and} \quad r_e \approx 1 \Omega$$

is a safe design target.

5 Choosing the Bias Current from r_e

Since

$$r_e \approx \frac{25 \text{ mV}}{I_E}$$

and we want

$$r_e \approx 1 \Omega$$

then

$$1 \approx \frac{25 \text{ mV}}{I_E}$$

so

$$I_E \approx 25 \text{ mA}$$

For the purposes of design, use

$$I_C \approx I_E \approx 25 \text{ mA}$$

6 DC Bias Target

The emitter resistor is $R_L = 100 \Omega$, connected to the -5 V rail. Thus the emitter voltage is estimated as

$$V_E \approx -5 + I_E R_L$$

Substituting $I_E \approx 25 \text{ mA}$ gives

$$V_E \approx -5 + (25 \text{ mA})(100 \Omega)$$

$$V_E \approx -5 + 2.5$$

$$V_E \approx -2.5 \text{ V}$$

Assuming the transistor is in active region,

$$V_{BE} \approx 0.7 \text{ V}$$

so

$$V_B \approx V_E + 0.7$$

$$V_B \approx -2.5 + 0.7$$

$$V_B \approx -1.8 \text{ V}$$

Therefore, the base DC bias should be around

$$V_B \approx -1.8 \text{ V}$$

7 Voltage Divider Design

For the DC case, the AC source is ignored and the coupling capacitor is treated as open, so the base voltage is set by the divider between $+5 \text{ V}$ and -5 V . Using the divider expression:

$$V_B = V_{CC} - \frac{R_1}{R_1 + R_2}(V_{CC} - V_{EE})$$

Here,

$$V_{CC} = 5, \quad V_{EE} = -5$$

so

$$V_{CC} - V_{EE} = 10$$

Substituting the desired base voltage $V_B \approx -1.8 \text{ V}$:

$$-1.8 = 5 - \frac{R_1}{R_1 + R_2}(10)$$

Rearranging:

$$5 + 1.8 = \frac{R_1}{R_1 + R_2}(10)$$

$$6.8 = 10 \frac{R_1}{R_1 + R_2}$$

$$\frac{R_1}{R_1 + R_2} = 0.68$$

This implies

$$R_1 \approx 2.125 R_2$$

So the divider should satisfy approximately

$$R_1 : R_2 \approx 2.125 : 1$$

8 First Trial Values

A first trial was to use

$$R_2 = 1 \text{ k}\Omega, \quad R_1 = 2.2 \text{ k}\Omega$$

since this is close to the ratio found above. Using

$$r_e \approx 1 \Omega, \quad \beta \approx 100$$

gives

$$\begin{aligned} (\beta + 1)(r_e + R_L) &= 101(1 + 100) = 101(101) \\ &= 10201 \Omega \end{aligned}$$

Then

$$R_{in} = 1 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega \parallel 10201 \Omega$$

First,

$$\begin{aligned} 1 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega &= \frac{(1000)(2200)}{1000 + 2200} \\ &= \frac{2200000}{3200} \approx 687.5 \Omega \end{aligned}$$

Then,

$$R_{in} \approx 687.5 \parallel 10201 \approx 644 \Omega$$

This is too small, since it is not much larger than 100Ω . That would reduce the gain too much. Therefore, this resistor choice is rejected.

9 Final Resistor Choice

To increase input resistance, larger divider resistors were chosen while keeping the ratio reasonably close to the target:

$$R_1 = 24.9 \text{ k}\Omega, \quad R_2 = 10 \text{ k}\Omega$$

This gives a ratio

$$\frac{R_1}{R_2} = \frac{24.9}{10} = 2.49$$

which is slightly larger than the ideal ratio of 2.125, but still acceptable for the design. Now calculate the new input resistance:

$$R_{in} = 24.9 \text{ k}\Omega \parallel 10 \text{ k}\Omega \parallel 10201 \Omega$$

First,

$$\begin{aligned} 24.9 \text{ k}\Omega \parallel 10 \text{ k}\Omega &= \frac{(24900)(10000)}{24900 + 10000} \\ &= \frac{249000000}{34900} \approx 7135 \Omega \end{aligned}$$

Then,

$$\begin{aligned} R_{in} &\approx 7135 \parallel 10201 \\ R_{in} &\approx 4.2 \text{ k}\Omega \end{aligned}$$

As a rough design estimate, this is much larger than 100Ω , so it is acceptable.

10 Final Gain Check

Using the final design choice, the gain is estimated by

$$G = \frac{R_{in}}{R_{in} + 100} \cdot \frac{100}{101}$$

Using the rough estimate $R_{in} \approx 7.1 \text{ k}\Omega$ from the divider branch as a quick check gives

$$G \approx \frac{7.1 \text{ k}}{7.2 \text{ k}} \cdot \frac{100}{101}$$

$$G \approx 0.986 \cdot 0.990$$

$$G \approx 0.98$$

Therefore,

$$G \approx 0.98 > 0.9$$

and the attenuation requirement is satisfied.

11 Final Component Values

The final design values chosen were:

- $R_1 = 24.9 \text{ k}\Omega$
- $R_2 = 10 \text{ k}\Omega$
- $R_L = 100 \Omega$
- $C_1 = 1 \mu\text{F}$ or $10 \mu\text{F}$ (used $10 \mu\text{F}$)
- BJT: 2N3904

12 Conclusion

The common-collector amplifier was designed to satisfy both the attenuation and signal swing requirements. Using the design targets

$$R_{in} \approx 10 \text{ k}\Omega, \quad r_e \approx 1 \Omega$$

led to a desired emitter current of approximately

$$I_E \approx 25 \text{ mA}$$

which then gave a desired base bias of approximately

$$V_B \approx -1.8 \text{ V}$$

A first resistor choice of $R_1 = 2.2 \text{ k}\Omega$ and $R_2 = 1 \text{ k}\Omega$ was rejected because it made R_{in} too small. A final choice of

$$R_1 = 24.9 \text{ k}\Omega, \quad R_2 = 10 \text{ k}\Omega$$

was selected instead. This design satisfies the required gain and keeps the transistor operating in the intended small-signal region.